

Practical - 4

Aim: Utilize core (http, fs, url, util, path, os, events) and local modules to demonstrate their output

1: Write a node.js code to establish connection with MySQL and create database “Node_test”.

Code:

```
const mysql = require('mysql2');

// Connect to MySQL
const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
});

con.connect((err) => {
  if (err) throw err;

  console.log('Connected to MySQL!');

  // Create database
  con.query('CREATE DATABASE IF NOT EXISTS Node_test', (err) => {
    if (err) throw err;

    console.log("Database 'Node_test' created successfully.");
    con.end();
  });
});
```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_1.js
Connected to MySQL!
Database 'Node_test' created successfully.
```

2: Write a node.js code to create a table ‘Product’ with Id, Name, Brand, Quantity and Price as fields.

Code:

```
const mysql = require('mysql2');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
```

```
});

con.connect((err) => {
  if (err) throw err;

  console.log('Connected to Node_test database!');
  const sql = `
    CREATE TABLE IF NOT EXISTS Product (
      Id INT AUTO_INCREMENT PRIMARY KEY,
      Name VARCHAR(100),
      Brand VARCHAR(100),
      Quantity INT,
      Price DECIMAL(10,2)
    )
  `;
  con.query(sql, (err) => {
    if (err) throw err;
    console.log('Product table created successfully.');
```

```
con.end();
  });
});
```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_2.js
Connected to Node_test database!
Product table created successfully.
```

3: Write a node.js program that takes product information like Name, Brand, Quantity and Price from the user using HTML form and stores it into the Product table as mentioned below:

Product				
Id	Name	Brand	Quantity	Price
1	Soap	Dettol	3	20
2	Shampoo	Loreal	5	100
3	Toothpaste	Colgate	2	50
4	Sanitizer	LifeBoy	2	60

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');
```

```
const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);

  // Display form
  if (url.pathname === '/') {
    res.writeHead(200, { 'Content-Type': 'text/html' });

    res.end(`
      <h1>Add Product</h1>
      <form action="/add" method="GET">
        Name:
        <input type="text" name="name" required>
        <br><br>
        Brand:
        <input type="text" name="brand" required>
        <br><br>
        Quantity:
        <input type="number" name="quantity" required>
        <br><br>
        Price:
        <input type="number" name="price" required>
        <br><br>
        <button type="submit">Add Product</button>
      </form>
    `);
  }

  // Insert product
  else if (url.pathname === '/add') {
    const name = url.searchParams.get('name');
    const brand = url.searchParams.get('brand');
    const quantity = url.searchParams.get('quantity');
    const price = url.searchParams.get('price');

    const sql = `
      INSERT INTO Product (Name, Brand, Quantity, Price)
      VALUES (?, ?, ?, ?)
    `;

    con.query(sql, [name, brand, quantity, price], (err) => {
      if (err) {
        res.writeHead(500);
        res.end('Database Error');
      }
    });
  }
});
```

```
        return;
    }
    res.writeHead(200, {
        'Content-Type': 'text/html',
    });
    res.end(`
        <h2>Product added successfully!</h2>
        <a href="/">Add another product</a>
    `);
});
}
});

server.listen(3000, () => {
    console.log('Server running at http://localhost:3000');
});
```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_3.js
Server running at http://localhost:3000
```

The first screenshot shows a web browser at localhost:3000 with the title "Add Product". It contains a form with four input fields: "Name:" with "Soap", "Brand:" with "Dettol", "Quantity:" with "3", and "Price:" with "20". Below the fields is an "Add Product" button.

The second screenshot shows the browser at localhost:3000/add?name=Soap&brand=Dettol&quantity=3&price=20. The page displays "Product added successfully!" in a large, bold font, followed by a blue link "Add another product".

4: Write a node.js program to retrieve all the details of the product and display in browser.

Code:

```
const http = require('http');
const mysql = require('mysql2');

const con = mysql.createConnection({
    host: 'localhost',
    user: 'vatsal',
    password: 'vatsal',
```

```

    database: 'Node_test',
  });
  const server = http.createServer((req, res) => {
    if (req.url === '/products') {
      const sql = 'SELECT * FROM Product';
      con.query(sql, (err, result) => {
        if (err) {
          res.writeHead(500);
          res.end('Database Error');
          return;
        }
        let html = `
          <html>
          <head>
            <title>Products</title>
            <style>
              table {
                border-collapse: collapse;
                width: 70%;
              }
              th, td {
                border: 1px solid black;
                padding: 10px;
                text-align: center;
              }
              th {
                background-color: lightgray;
              }
            </style>
          </head>
          <body>
            <h1>Product Details</h1>
            <table>
              <tr>
                <th>Id</th>
                <th>Name</th>
                <th>Brand</th>
                <th>Quantity</th>
                <th>Price</th>
              </tr>
            </table>
          `;

        result.forEach((product) => {
          html += `
            <tr>
              <td>${product.Id}</td>
              <td>${product.Name}</td>
              <td>${product.Brand}</td>
              <td>${product.Quantity}</td>
              <td>${product.Price}</td>
            </tr>
          `;
        });
      });
    }
  });

```

```

        </tr>
      `;
    });
    html += `
      </table>
    </body>
  </html>
  `;
  res.writeHead(200, {
    'Content-Type': 'text/html',
  });
  res.end(html);
});
} else {
  res.writeHead(404);
  res.end('404 - Page Not Found');
}
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000/products');
});

```

Output:

```

C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_4.js
Server running at http://localhost:3000/products

```

← → ↻ http://localhost:3000/products

Product Details

Id	Name	Brand	Quantity	Price
1	Soap	Dettol	3	20.00
2	Shampoo	Loreal	5	100.00
3	Toothpaste	Colgate	2	50.00
4	Sanitizer	LifeBoy	2	60.00

5: Write a node.js program to search any product by Name or Brand.

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);
  if (url.pathname === '/') {
    res.writeHead(200, {
      'Content-Type': 'text/html',
    });
    res.end(`
      <h1>Search Product</h1>
      <form action="/search" method="GET">
        Enter Name or Brand:
        <input type="text" name="value" required>
        <button type="submit">Search</button>
      </form>
    `);
  } else if (url.pathname === '/search') {
    const value = url.searchParams.get('value');
    const sql = `
      SELECT * FROM Product
      WHERE Name LIKE ? OR Brand LIKE ?
    `;
    const searchValue = `%${value}%`;
    con.query(sql, [searchValue, searchValue], (err, result) => {
      if (err) {
        res.writeHead(500);
        res.end('Database Error');
        return;
      }
      let html = `
        <h1>Search Results</h1>
        <table border="1" cellpadding="10">
          <tr>
            <th>Id</th>
            <th>Name</th>
            <th>Brand</th>
            <th>Quantity</th>
            <th>Price</th>
          </tr>
      `;
    });
  }
});
```

```
`;  
result.forEach((product) => {  
  html += `  
    <tr>  
      <td>${product.Id}</td>  
      <td>${product.Name}</td>  
      <td>${product.Brand}</td>  
      <td>${product.Quantity}</td>  
      <td>${product.Price}</td>  
    </tr>  
  `;  
});  
html += '</table>';  
res.writeHead(200, {  
  'Content-Type': 'text/html',  
});  
res.end(html);  
});  
} else {  
  res.writeHead(404);  
  res.end('404 - Page Not Found');  
}  
});  
server.listen(3000, () => {  
  console.log('Server running at http://localhost:3000');  
});
```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_5.js  
Server running at http://localhost:3000
```

← → ↻ http://localhost:3000/

Search Product

Enter Name or Brand:

← → ↻ http://localhost:3000/search?value=Loreal

Search Results

Id	Name	Brand	Quantity	Price
2	Shampoo	Loreal	5	100.00

6: Write a node.js program to retrieve all the products whose quantity is less than 3.

Code A:

```
const http = require('http');
const mysql = require('mysql2');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  if (req.url === '/products') {
    const sql = 'SELECT * FROM Product WHERE Quantity < 3';
    con.query(sql, (err, result) => {
      if (err) {
        res.writeHead(500);
        res.end('Database Error');
        return;
      }
      let html = `
        <h1>Products With Quantity Less Than 3</h1>
        <table border="1" cellpadding="10">
          <tr>
            <th>Id</th>
            <th>Name</th>
            <th>Brand</th>
            <th>Quantity</th>
            <th>Price</th>
          </tr>
      `;
      result.forEach((product) => {
        html += `
          <tr>
            <td>${product.Id}</td>
            <td>${product.Name}</td>
            <td>${product.Brand}</td>
            <td>${product.Quantity}</td>
            <td>${product.Price}</td>
          </tr>
        `;
      });
      html += '</table>';
      res.writeHead(200, {
        'Content-Type': 'text/html',
      });
      res.end(html);
    });
  } else {
```

```
    res.writeHead(404);
    res.end('404 - Page Not Found');
  }
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000/products');
});
```

Output A:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_6.js
Server running at http://localhost:3000/products
```

← → ↻ http://localhost:3000/products

Products With Quantity Less Than 3

Id	Name	Brand	Quantity	Price
3	Toothpaste	Colgate	2	50.00
4	Sanitizer	LifeBoy	2	60.00

7: Write a node.js program to update price of a specific item in the product table.

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);
  // Display form
  if (url.pathname === '/') {
    res.writeHead(200, {
      'Content-Type': 'text/html',
```

```

});
res.end(`
  <h1>Update Product Price</h1>
  <form action="/update" method="GET">
    Product ID:
    <input type="number" name="id" required>
    <br><br>
    New Price:
    <input type="number" name="price" required>
    <br><br>
    <button type="submit">Update Price</button>
  </form>
`);
}
// Update price
else if (url.pathname === '/update') {
  const id = url.searchParams.get('id');
  const price = url.searchParams.get('price');

  const sql = `
    UPDATE Product
    SET Price = ?
    WHERE Id = ?
  `;

  con.query(sql, [price, id], (err, result) => {
    if (err) {
      res.writeHead(500);
      res.end('Database Error');
      return;
    }
    res.writeHead(200, {
      'Content-Type': 'text/html',
    });
    res.end(`
      <h2>Product price updated successfully.</h2>
      <p>Product ID: ${id}</p>
      <p>New Price: ${price}</p>

      <a href="/">Go Back</a>
    `);
  });
} else {
  res.writeHead(404);
  res.end('404 - Page Not Found');
}
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000');
});

```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_7.js
Server running at http://localhost:3000
```

← → ↻ http://localhost:3000/

Update Product Price

Product ID:

New Price:

← → ↻ http://localhost:3000/update?id=3&price=70

Product price updated successfully.

Product ID: 3

New Price: 70

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8: Write a node.js program to delete specific product from the product table.

Code:

```
const http = require('http');
const mysql = require('mysql2');
const { URL } = require('url');

const con = mysql.createConnection({
  host: 'localhost',
  user: 'vatsal',
  password: 'vatsal',
  database: 'Node_test',
});

const server = http.createServer((req, res) => {
  const url = new URL(req.url, `http://${req.headers.host}`);

  // Display form
  if (url.pathname === '/') {
    res.writeHead(200, {
```

```

    'Content-Type': 'text/html',
  });

  res.end(`
    <h1>Delete Product</h1>

    <form action="/delete" method="GET">

      Product ID:
      <input type="number" name="id" required>

      <button type="submit">Delete</button>

    </form>
  `);
}

// Delete product
else if (url.pathname === '/delete') {
  const id = url.searchParams.get('id');

  const sql = 'DELETE FROM Product WHERE Id = ?';
  con.query(sql, [id], (err, result) => {
    if (err) {
      res.writeHead(500);
      res.end('Database Error');
      return;
    }
    res.writeHead(200, {
      'Content-Type': 'text/html',
    });
    res.end(`
      <h2>Product deleted successfully.</h2>

      <p>Deleted Product ID: ${id}</p>

      <a href="/">Go Back</a>
    `);
  });
} else {
  res.writeHead(404);
  res.end('404 - Page Not Found');
}
});

server.listen(3000, () => {
  console.log('Server running at http://localhost:3000');
});

```

Output:

```
C:\Users\Kavya\Desktop\temp\SP_Practicals\Practicals\Practical-4>node Practical_8.js  
Server running at http://localhost:3000
```

← → ↻ http://localhost:3000/

Delete Product

Product ID:

← → ↻ http://localhost:3000/delete?id=4

Product deleted successfully.

Deleted Product ID: 4

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